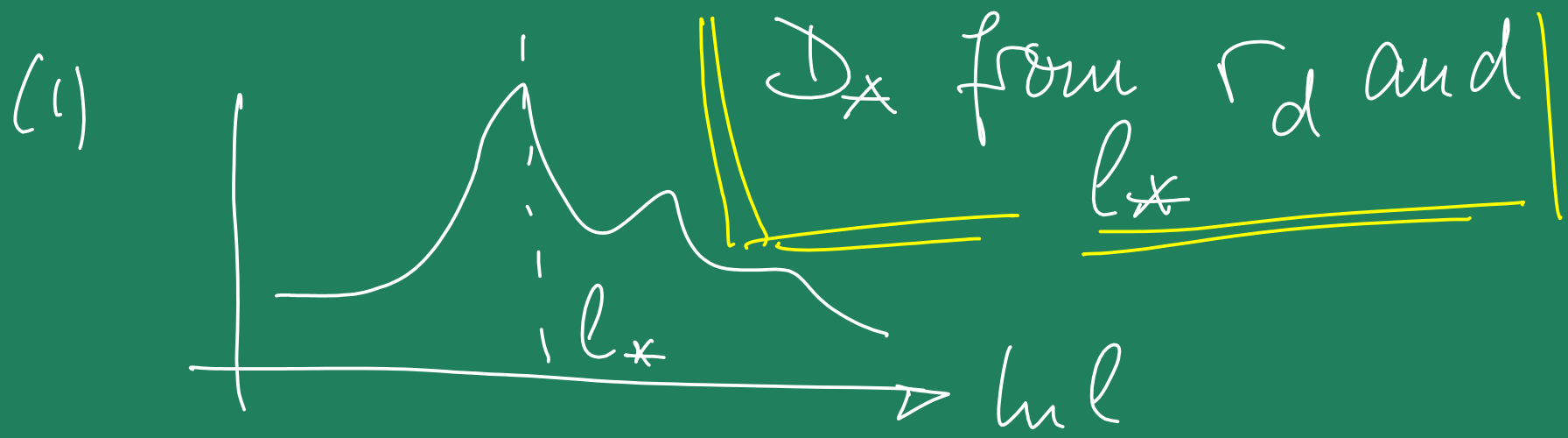




(4) BAO: $r_d = 143.4 \text{ Mpc}$

$$\Delta\mu \approx 5 \log(1.434 \cdot 10^7)$$

$$\approx \underline{\underline{35.8}}$$

(2)

$$\frac{r_d}{D_A} \approx \theta ; \quad \theta = \frac{\pi}{l}$$

$$\frac{\pi}{l_*} \approx \frac{r_d}{D_A} ; \quad D_A = \frac{l_*}{\pi} r_d$$

(3)

$$\mu = 5 \log \frac{DL}{10 \text{ pc}} = 5 \log \left(\frac{D_A}{a 10 \text{ pc}} \right)$$

$$= \underbrace{5 \log \left(\frac{l_*}{\pi a} \right)}_{\mu_*} + \underbrace{5 \log \left(\frac{r_d}{10 \text{ pc}} \right)}_{\Delta\mu}$$

$$\mu_* \approx 5 \log \left(\frac{200}{\pi} 1100 \right)$$

$$\approx 24.2$$

$$\underline{\underline{\mu_{\text{CMB}} \approx 60}}$$

D_A estimate

$$\begin{aligned} W_{\text{CMB}} &= \int_{a_*}^1 \frac{c da}{a \dot{a}} = \frac{c}{H_0} \int_{a_*}^1 \frac{da}{a^2 E(a)} \stackrel{(EdS)}{\approx} \frac{c}{H_0} \int_{a_*}^1 \frac{da}{\sqrt{\Omega_{m0}} \sqrt{a}} \\ &= \frac{2c}{H_0 \sqrt{\Omega_{m0}}} (1 - \sqrt{a_*}) \approx \frac{2c}{H_0 \sqrt{\Omega_{m0}}} \approx 11 \text{ Gpc} \end{aligned}$$

$$D_A \approx a_* W_{\text{CMB}} \approx 10^{-2} \text{ Gpc} \approx 10 \text{ Mpc}$$

$$D_L \approx \frac{W_{\text{CMB}}}{a_*} \approx 1.21 \cdot 10^{13} \text{ pc}$$

$$\underline{\underline{\mu}} = 5 \log\left(\frac{D_L}{10 \text{ pc}}\right) \approx 5 \log(1.21 \cdot 10^{12}) \approx \underline{\underline{60.4}}$$